MILITARY SPECIFICATION

CIRCUIT BREAKER, AIRCRAFT, TRIP-FREE
PUSH PULL, TYPE I, DUAL SAFETY, -55 THRU +121°C

This specification is approved for use by all departments and agencies of the Department of Defense.

The complete requirements for procuring the circuit breaker described herein shall consist of this document and the latest issue of MIL-C-5809.

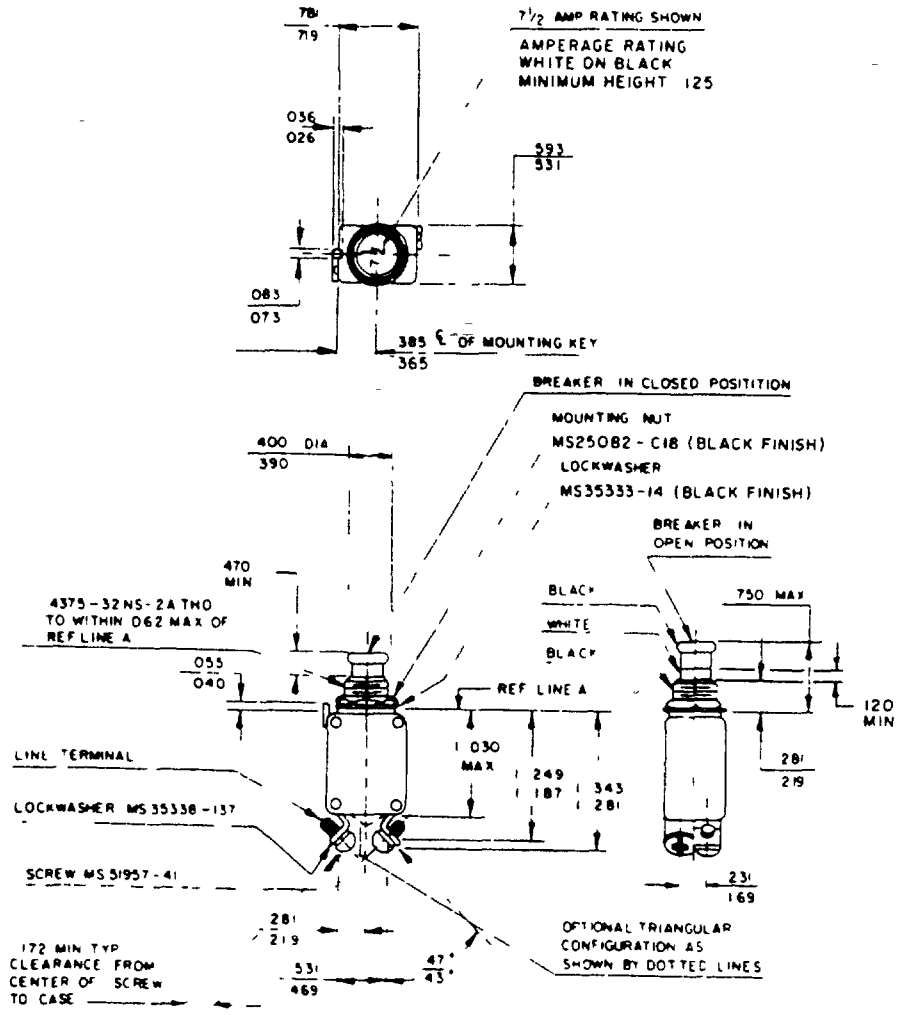


FIGURE 1. Circuit breaker dimensions.

AMSC N/A FSC 5925

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TABLE I

Dash No. See Req 1	Nominal Amperage Rating (A)	Max. Volt Drop	Weight Max. Lbs.	Detail Calibration Performance								
				Ult. Trip 25°C		Overload Cal. Time in Secs. -55° to +121°C			Ambient Effect on Calibration -121°C		-55°C	
				Min.	Max.	200%	500%	1000%	Ult. Trip Min.	Ult. Trip Max.	Ult. Trip Min.	Ult. Trip Max.
2-1/2	2-1/2	.70	.055	115	138	1.5 to 40	0.15 to 3.0	0.38 to 0.8	90	138	115	165
3	3	.55										
5	5	.350										
7-1/2	7-1/2	.300										
10	10	.280										

1/ Min and max ultimate trip expressed in % of rated current testing. Time one hour.

TABLE II

Nominal Amperage Rating (A)	Interrupting Capacity - Amperes						Endurance-Cycles				
							Resistive		Inductive		Mech No. Load
	A	B	C	D	E	F	AC	DC	AC	DC	
2-1/2	2800	2800	6000	6000	(a) 1500 (b) 750	(a) 3000 (b) 1000	5000	5000	2500	2500	10,000
3	2500	2500									
5											
7-1/2											
10											

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TABLE III

Nominal Amperage Rating (A)	Operating Force Lbs.	
	Pullout	Reset
2-1/2	.75 to 5	1.2 to 8
3		
5		
7-1/2		
10		

TABLE IV. Circuit breaker link separation characteristics.

Nominal Amperage Rating (A)	Maximum Circuit Breaker Link Separation Times in Seconds As a Function of Current Overload						
	400%	500%	600%	700%	800%	900%	1000%
2-1/2	--	--	34.0	20.0	13.0	9.0	6.0
3	--	--	34.0	20.0	13.0	9.0	6.0
5	--	95.0	36.0	18.0	10.0	6.0	3.5
7-1/2	69.0	28.0	14.0	8.0	4.0	3.5	2.0
10	60.0	35.0	20.0	12.0	7.0	4.0	2.5

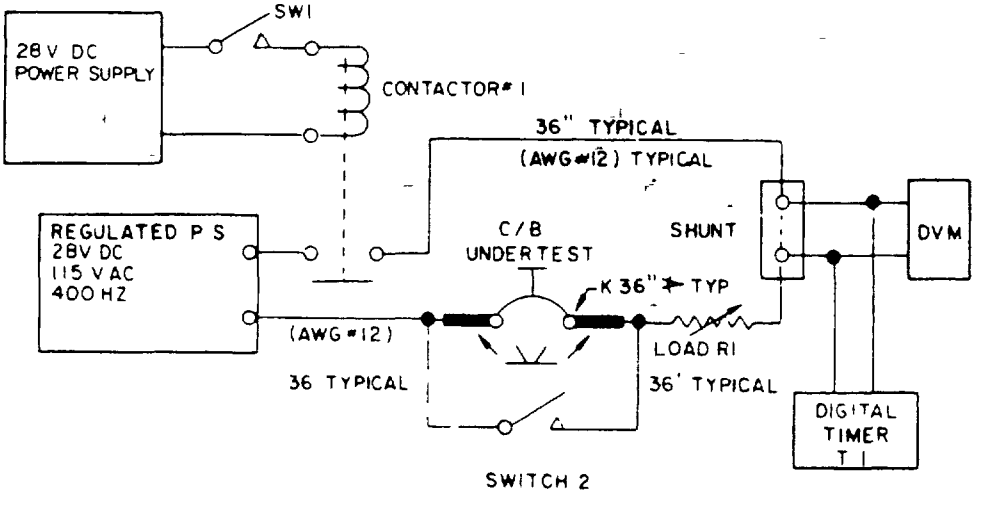


FIGURE 2. Fusing characteristics test circuit.

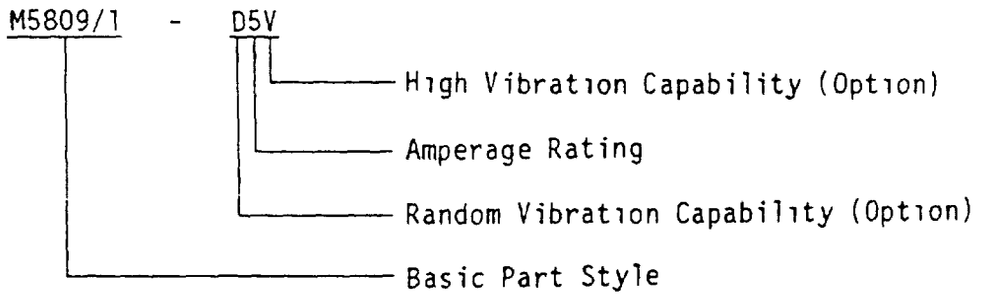
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TABLE V. Environmental performance.

Max Operating Altitude -	70,000 feet
Operating Ambient Temp Range	-55°C to +121°C
Vibration - In accordance with MIL-C-5809	Sine - Required High Level Sine - (Optional) Random - (Optional)
Shock	50G, MIL-STD-202, Method 213, Condition A
Acceleration	10G

REQUIREMENTS:

1. The part numbers for circuit breakers in accordance with this specification shall conform to the example below.



When a designator is not applicable it shall be omitted from the part number.

2. Circuit breaker link separation characteristics: When tested in accordance with Requirement 3, the circuit breaker separable link shall open in the times shown in Table IV. The leakage current between the line and load terminals shall be less than 1 milliamp at 900 VAC with the breaker in the closed or reset position.

3. Circuit breakers shall be subjected to the separable link characteristic test currents of Table II. The normal trip mechanism shall be held closed to prevent tripping electrically without applying undue stress on the current sensing mechanism or altering the circuit breakers thermal dissipation characteristics. The technique may be vendor specific. Circuit breakers shall be mounted on 1/8" thick aluminum panels and tested using the circuit from Figure 2 as follows:

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1. Shut contactors 1 and 2 with circuit breaker open.

2. Adjust current and voltage to test levels with the regulated power supply and load resistor R1.

3. Open circuit by opening contactors 1 and 2.

4. Close circuit breaker under test and shut contactor 1.

5. Fine adjust current with R1 and record link separation time.

6. Open contactor 1.

One separable link test will be performed on each of the following circuit breaker samples. Failure of the breaker to open within the maximum time specified shall constitute failure of the breaker to pass the test.

TABLE VI. Fusing characteristic test (sample size vs overload).

Nominal Amperage Rating (A)	Number of Breakers to be Subjected to Given Current Overload							Total Samples Required
	400%	500%	600%	700%	800%	900%	1000%	
2-1/2	--	--	7	7	5	3	3	26
3.0	--	--	7	7	5	3	3	26
5.0	--	7	7	7	3	3	3	30
7-1/2	6	6	6	6	5	3	3	35
10.0	6	6	6	6	5	3	3	35

NOTES.

1. This device exhibits all the standard performance characteristics of aircraft, trip-free, push pull Type I circuit breakers as well as providing an additional "dual safety" backup feature in which an independent separable link will clear overloads should the normal tripping mechanism be disabled. Emperical testing has shown that this device will operate in sufficient time to protect wire types MIL-W-22759/32 and MIL-W-81381/21 and /22 wire types. Suitability for use with any other wire type has not been verified by actual testing.

2. Nominal voltage ratings -28V DC or 115V AC 400 Hz.

3. Dimensions are in inches unless otherwise specified. Tolerances: Decimals ±.031, unless otherwise specified.

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4. All numerical values listed in percentage in Tables I and IV are expressed in percent of rated current for each amperage rating.

Custodians:
Navy - AS
Air Force - 85
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Preparing Activity.
Navy - AS
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